

xiong-et-al-2020-the-curse-of-knowledge-in-visual-data-co

Auto-assembled report. Section content is drawn from the summarization pipeline outputs.

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Paper Synthesis

Not available for this paper.

Abstract

Preserved from source without summarization.

Abstract--A viewer can extract many potential patterns from any set of visualized data values. But that means that two people can see different patterns in the same visualization, potentially leading to miscommunication. Here, we show that when people are primed to see one pattern in the data as visually salient, they believe that naïve viewers will experience the same visual salience. Participants were told one of multiple backstories about political events that affected public polling data, before viewing a graph that depicted those data. One pattern in the data was particularly visually salient to them given the backstory that they heard. They then predicted what naïve viewers would most visually salient on the visualization. They were strongly influenced by their own knowledge, despite explicit instructions to ignore it, predicting that others would find the same patterns to be most visually salient. This result reflects a psychological phenomenon known as the curse of knowledge, where an expert struggles to re-create the state of mind of a novice. The present findings show that the curse of knowledge also plagues the visual perception of data, explaining why people can fail to connect with audiences when they communicate patterns in data.

Introduction (Paper-Level)

Not available for this paper.

General Discussion

Not available for this paper.

Critique

Not available for this paper.